

The Value of Slack: Capacity Utilization in Multi-Product Firms*

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Abstract

This paper studies how multi-product firms utilize capacity when product demand is uncertain. Using novel *product-level* data on manufacturing firms, we document four facts. Capacity utilization varies substantially across products within the same firm; products with more volatile demand are used less intensively; capacity and sales shares are strongly related to average demand; and common firm-level shocks have different implications for capacity utilization than idiosyncratic product-level shocks. We rationalize these patterns in a tractable structural model of multi-product capacity choice under demand uncertainty. It shows that firm-level volatility is not sufficient to characterize firms' exposure to demand risk. Two firms with identical observable firm-level volatility may allocate capacity and earn profits in different ways depending on whether shocks are correlated across product lines or idiosyncratic to individual products. Consequently, policies that alter demand risk or capacity costs impact the economy not just through firm scale, but by reshaping the internal allocation of capacity across product lines.

Keywords: .

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1 Introduction

Firms in modern economies routinely operate well below full capacity. In US manufacturing, average industrial capacity utilization has rarely exceeded $[\bar{u}^{US}]$ %. In Indian manufacturing, average product-level utilization is around $[X]$ %, with a range of $[Y]$ – $[Z]$ % across products within a firm.

This persistent underutilization is not noise: it reflects deliberate choices made under demand uncertainty. Yet the existing literature has approached this phenomenon from two parallel angles that fail to meet. The *capacity utilization* literature treats utilization as a single firm- or plant-level margin and finds that idiosyncratic, not aggregate, forces explain most of the variation. The *multi-product firm* literature studies the extensive margin—which products firms make (Bernard, Redding, and Schott, 2010a)—but is essentially silent on the intensive margin of how a firm divides shared capacity across the product lines it has chosen to operate. This paper sits at this missing intersection: *taking the firm’s product portfolio as given, we ask how firms allocate capacity across product lines when products face different degrees of demand uncertainty.*

To answer this question, we combine novel product-level data on installed capacity and physical output from Indian manufacturing with a tractable structural model of multi-product capacity allocation under demand uncertainty. The empirical patterns and the model together deliver a central insight: two firms with identical observable firm-level demand volatility allocate capacity and earn profits in fundamentally different ways depending on whether their shocks are correlated or idiosyncratic across product lines—a within-firm distinction that is invisible to firm-level data but is exactly what is required to answer policy and aggregation questions about demand-risk exposure.

Literature Review.

Outline. The rest of the article is organized as follows. Section ?? describes data and measurement. Section ?? describes a theoretical model. Section ?? lays out the estimation strategy. Section ?? presents the main results. Section ?? concludes.

2 Data

The following paragraphs detail the source and construction of each dataset.

Firm Financials. We obtain firm- and product-level information from Prowess, a database compiled by the Centre for Monitoring the Indian Economy (CMIE). Prowess has been widely used in studies of Indian firms, including [Bertrand et al. \(2002\)](#); [Gopalan et al. \(2007\)](#); [Lilienfeld-Toal et al. \(2012\)](#); [Gopalan et al. \(2016\)](#); [Naaraayanan and Nielsen \(2021\)](#); and [Naaraayanan and Wolfenzon \(2024\)](#). We use the most recent vintage of the database, which addresses the systematic survivorship bias present in earlier versions, as discussed by [Siegel and Choudhury \(2012\)](#).

CMIE collects information from balance sheets and income statements for roughly 37,000 publicly listed and private firms. Firms in Prowess account for more than 70% of industrial output, 75% of corporate taxes, and over 95% of excise taxes collected by the Government of India, and primarily represent large and medium-sized firms in the economy ([Bau and Matray, 2023](#)).¹ These features make the data especially well suited for studying how firms adjust capacity utilization over time in response to demand uncertainty.

Product-level Capacity Utilization. To examine within-firm capacity utilization, we use detailed product-level information that firms disclose under the reporting requirements of the Companies Act of 1956 and the Companies Act of 2013. The data report total sales and total quantity sold at the firm-product level, which allows us to construct unit prices and quantities. Importantly, the data also contain information on installed capacity, actual production, and sales reported in company annual reports; see [Goldberg et al. \(2010\)](#) and [Bau and Matray \(2023\)](#) for further details.²

Our final dataset contains approximately 2,300 distinct manufactured products, where product definitions are held fixed over time. We use Prowess’s internal product classification, which is based on the National Industrial Classification (NIC). With these definitions, we construct a panel of product-level output and prices, defining the unit price of each product as total sales divided by total quantity sold.³

¹Prior research on India has relied on the Annual Survey of Industries (ASI) to study the manufacturing sector. A key distinction is that Prowess is a firm-level panel dataset, while ASI is an establishment-level dataset based on a repeated cross-section of roughly 30,000 establishments per year ([Martin, 2011](#); [Sivadasan, 2009](#)). The limited panel dimension and the lack of information on product-level capacity utilization in ASI makes it unsuitable for studying determinants of capacity utilization.

²The Companies Act does not require firms to report products in standardized units. As a result, units may vary across firms and within firms over time. We therefore standardize units within and across firms and drop observations when the available information is insufficient to reconcile changes in unit types within a firm-product over time.

³[Figure IAA1](#) and [Figure IAA2](#) report the distribution of products and the share of total manufacturing output accounted for by each two-digit manufacturing sector. Basic metals, vehicles, and chemicals account for the largest shares of value, while furniture and other wood products account for the smallest. These patterns are consistent with the product distributions documented for samples of multi-product firms in the United States ([Bernard et al., 2010b](#)) and India ([De Loecker et al., 2016](#)).

For each product, we define *Utilization* as the ratio of actual output to installed capacity at the firm-product-year level. Table 1 reports descriptive statistics for the final estimation sample. The data reveal substantial heterogeneity across firms. Firm size, either measured by assets or sales, is highly right-skewed with the sample containing a small number of very large firms. The average firm has a leverage ratio of 0.24 and produces about three distinct products. Export participation, however, is highly concentrated. The median firm earns only 0.70% of revenues from exports, while firms at the 90th percentile earn more than 42% of revenues from foreign sales.

Panel B reports firm-product-level characteristics. Capacity utilization averages 0.57, but dispersion is large, with a standard deviation of 0.37. At the 10th percentile, utilization is only 0.07, indicating substantial slack in a nontrivial share of product lines. Revenue shares are also highly concentrated within firms. The median product accounts for 58% of firm revenues, while the 75th percentile product accounts for 99%. By the 90th percentile, the product effectively accounts for all firm revenues. Thus, although multi-product firms are common in the data, firm performance is often dominated by a single core product, with remaining product lines accounting for relatively small revenue shares.

TABLE 1: Summary statistics

This table summarizes the sample used for the analysis. The sample keeps product-year observations with nonmissing capacity utilization, product-level sales, proportion of firm sales, and the firm characteristics listed in the table. The time period spans the fiscal years between 1995 to 2016. In our analysis, “year” refers to the fiscal year as opposed to the calendar year because the financial year in India runs from April 1 to March 31. Product characteristics are measured at the product-year level, so they have the same number of observations. Firm characteristics are measured after reducing this same product-year subsample to one observation per firm-year. Capacity utilization is winsorized at the 5th and 95th percentiles, consistent with the main analysis. Data source: CMIE Prowess.

Variable	N	Mean	SD	p10	p25	p50	p75	p90
<i>Panel A. Firm characteristics</i>								
Assets	34,856	3,675.68	24,053.50	68.60	149.50	421.70	1,463.70	5,173.60
Sales	23,482	2,703.29	12,776.06	50.20	148.10	447.55	1,496.90	4,852.40
Leverage	31,448	0.24	0.38	0.04	0.10	0.19	0.31	0.45
Exporting intensity	34,853	11.85	22.84	0.00	0.00	0.70	11.66	42.37
Number of product lines	34,860	3.08	2.53	1.00	1.00	2.00	4.00	6.00
Investment/assets	20,855	-0.10	0.18	-0.23	-0.13	-0.05	-0.02	-0.01
Market-to-book	5,510	0.80	1.15	0.08	0.17	0.39	0.94	2.05
<i>Panel B. Firm-product characteristics</i>								
Capacity utilization	66,931	0.57	0.37	0.07	0.26	0.56	0.83	1.05
Product-level sales	60,706	1,313.25	11,387.42	7.20	40.80	180.30	642.90	2,020.50
Proportion of firm sales	60,637	0.54	0.40	0.01	0.10	0.58	0.99	1.00

Validation. As a validation exercise, we compare the annual capacity-utilization series in our data with two external benchmarks for India. In panel A of [Figure IAA3](#), the first external benchmark is the Reserve Bank of India OBICUS series, which reports quarterly capacity utilization for Indian manufacturing firms; we aggregate that quarterly series to the annual level. In panel B of [Figure IAA3](#), the second external benchmark is the OECD capacity-utilization tendency series for India, available through FRED as ‘BSCUJG02INQ160S’; we likewise convert the quarterly observations to annual averages. To place the three series on a common scale, each series is standardized over the common sample period before plotting. The figure shows that the movements in our data line up closely with the external aggregate series, which supports the use of the product-based utilization measure as a proxy for economy-wide capacity utilization.

Quantity Productivity, Markups, and Marginal Costs. For our analysis, we estimate quantity-based productivity following [De Loecker, Goldberg, Khandelwal, and Pavcnik \(2016\)](#), which allows us to separate physical productivity from markups, marginal costs, and quality differences. The estimation uses single-product firms to recover product-specific production function parameters and then applies these estimates to multi-product firms under standard assumptions about common product technologies and selection into product scope. We account for unobserved quality differences by using input price variation, modeled as a function of output prices, product market shares, product fixed effects, location fixed effects, and export status. This procedure yields firm-product-year measures of quantity productivity, markups, and marginal costs that are adjusted for product quality and selection into multi-product production. See [Appendix B](#) for more details.

Product-level Demand Shocks. We construct product-level demand shocks using an instrumental variables approach that addresses the endogeneity of prices. Our identification strategy exploits variation in marginal costs as an instrument for prices, following [Di Giovanni, Levchenko, and Mejean \(2014\)](#).

Specifically, we estimate the following demand equation using two-stage least squares:

$$\ln Q_{ijt} = \eta \ln P_{ijt} + \alpha_{ij} + \delta_{jt} + \varepsilon_{ijt}^D \quad (1)$$

where Q_{ijt} is the quantity produced of product j by firm i in year t , P_{ijt} is the product price, α_{ij} are firm-product fixed effects, and δ_{jt} are product-year fixed effects. We instrument for $\ln P_{ijt}$ using log marginal cost $\ln MC_{ijt}$, which shifts supply but is excluded from the demand equation under standard assumptions.

The demand residual is then:

$$\hat{\varepsilon}_{ijt}^D = \ln Q_{ijt} - \hat{\eta} \ln P_{ijt} \quad (2)$$

where $\hat{\eta}$ is the IV estimate of the price elasticity.

We measure product-level demand volatility as the standard deviation of demand residuals over time for each firm-product pair:

$$\sigma_{ij}^D = \text{sd}(\hat{\varepsilon}_{ijt}^D) \quad (3)$$

This captures the unpredictable component of demand facing each product line, purged of systematic price effects and common product-year shocks.

To distinguish between shocks that affect all products within a firm versus product-specific shocks, we decompose demand fluctuations as follows. First, we demean log product sales by firm-product:

$$\tilde{D}_{ijt} = \ln S_{ijt} - \overline{\ln S}_{ij} \quad (4)$$

where S_{ijt} is product sales and $\overline{\ln S}_{ij}$ is the firm-product mean.

We then decompose the demeaned sales into a firm-level component and an idiosyncratic component:

$$s_{it}^f = \frac{1}{J_{it}} \sum_{j \in \mathcal{J}_{it}} \tilde{D}_{ijt} \quad (5)$$

$$s_{ijt}^\varepsilon = \tilde{D}_{ijt} - s_{it}^f \quad (6)$$

where $\mathcal{J}_{it}$ is the set of products produced by firm i in year t , and $J_{it} = |\mathcal{J}_{it}|$.

The firm shock s_{it}^f captures demand movements common to all products within a firm-year—reflecting, for example, firm-wide demand conditions, aggregate shocks to the firm’s customer base, or macroeconomic factors affecting the firm. The idiosyncratic shock s_{ijt}^ε captures product-specific demand fluctuations orthogonal to the firm-level component.

In our empirical specifications, we use the absolute values $|s_{it}^f|$ and $|s_{ijt}^\varepsilon|$ to measure the magnitude of shocks, as both positive and negative deviations represent uncertainty that firms must manage through capacity decisions.

3 Empirical Facts

We organize the descriptive evidence around four empirical facts. The unit of observation is a firm-product-year, except where we explicitly aggregate to the firm-year level.

Fact 1. *Product-level utilization averages 56.9%, with within-firm range 48.8–66.2%. Firm fixed effects absorb approximately 38.3% of total dispersion; the remaining approximately 61.7% is within-firm across products.*

Table 2 shows that product-level capacity utilization displays substantial dispersion. This dispersion is not simply a comparison across large and small firms. When we absorb firm fixed effects, a meaningful share of the remaining variation continues to arise within firms, across product lines. In other words, firms do not operate all products at similar rates of capacity utilization. Instead, the same firm often combines highly utilized product lines with product lines that have substantial slack.

This fact is useful because it motivates a product-level analysis of capacity. A firm-level utilization measure would mask the heterogeneity in how capacity is used across products inside the firm. The relevant margin is therefore not only whether firms have idle capacity in the aggregate, but also how this slack is distributed across the firm’s product portfolio.

TABLE 2: Product-Level Capacity Utilization Dispersion

<i>Panel A: Summary Statistics</i>	
Mean product-level utilization	56.9%
Mean within-firm minimum	48.8%
Mean within-firm maximum	66.2%
<i>Panel B: Variance Decomposition</i>	
Share explained by firm fixed effects	38.3%
Within-firm share (across products)	61.7%

Estimating equation:

$$u_{ijt} = \alpha_i + \varepsilon_{ijt}$$

where u_{ijt} is capacity utilization for product j in firm i at time t , and α_i is a firm fixed effect. The variance decomposition reports the R^2 from regressing utilization on firm fixed effects. Sample includes firm-product-years with nonmissing capacity utilization. The time period spans the fiscal years between 1992 to 2016. In our analysis, “year” refers to the fiscal year as opposed to the calendar year because the financial year in India runs from April 1 to March 31.

Fact 2. *Product-level utilization declines with product-level demand volatility (-0.0360). Aggregating the same observations to the firm level, the volatility-utilization coefficient is -0.0319 and the explanatory power drops by 0.0039.*

In Table 3, we next relate product utilization to demand volatility. We estimate regressions of product-level utilization on product-level demand volatility, with year fixed effects

and standard errors clustered at the firm level. The coefficient on product demand volatility is negative and statistically significant. In the baseline specification, a one-unit increase in product-level volatility is associated with a 0.036 decline in product-level utilization. Thus, products that face more volatile demand are operated less intensively.

We also aggregate the same information to the firm-year level, weighting product-level utilization and volatility by product sales shares. The firm-level relationship between volatility and utilization remains negative, but the regression has less explanatory power. This attenuation is consistent with the idea that aggregation masks product-level heterogeneity. Firms can hold slack in some product lines while using other product lines more intensively, so the relationship between risk and utilization is better identified at the product level than in firm-level averages.

TABLE 3: Demand Volatility and Capacity Utilization

	Product-level (1)	Firm-level (2)
Demand volatility	-0.0360*** (0.0055)	-0.0319*** (0.0056)
R^2	0.0143	0.0104
Year fixed effects	Yes	Yes

Estimating equation:

$$u_{ijt} = \beta \cdot \sigma_{ij}^D + \delta_t + \varepsilon_{ijt}$$

where u_{ijt} is capacity utilization, σ_{ij}^D is product demand volatility (standard deviation of IV-based demand residuals), and δ_t are year fixed effects. Column (1) uses product-level observations; Column (2) aggregates to firm-level using sales-weighted averages. The time period spans the fiscal years between 1992 to 2016. In our analysis, “year” refers to the fiscal year as opposed to the calendar year because the financial year in India runs from April 1 to March 31. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Fact 3. *Firms’ product sales share is negatively correlated with product demand volatility. Firms also allocate smaller capacity shares to more volatile product lines.*

In Table 4, we then examine whether firms allocate more capacity and sales to products with different demand characteristics. The specification regresses a product’s capacity share (column 1), or sales share (column 2), on product demand volatility and average product demand, absorbing firm and product-year fixed effects. Firm fixed effects are important because they compare products within the same firm, holding fixed the firm’s overall scale, financial condition, and common demand environment. Both coefficients are negative and

economically significant. This suggests that firms’ within-year allocation of capacity and sales is more closely tied to the volatility of that demand.

TABLE 4: Capacity Allocation and Demand Volatility

	Capacity Share (1)	Sales Share (2)
Demand volatility	-0.0597** (0.0189)	-0.0401*** (0.0108)
R^2	0.80	0.84
Firm fixed effects	Yes	Yes
Product $\times$ year fixed effects	Yes	Yes

Estimating equation:

$$y_{ijt} = \beta \cdot \sigma_{ij}^D + \alpha_i + \delta_{jt} + \varepsilon_{ijt}$$

where y_{ijt} is either the capacity share or sales share of product j in firm i . Sales share is the proportion of firm sales from product j . Capacity share is the material cost share, recovered from production function estimates. σ_{ij}^D is product demand volatility, α_i are firm fixed effects, and δ_{jt} are product-by-year fixed effects. The time period spans the fiscal years between 1992 to 2016. In our analysis, “year” refers to the fiscal year as opposed to the calendar year because the financial year in India runs from April 1 to March 31. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Fact 4. *Decomposing demand volatility into firm-level and idiosyncratic components, utilization falls with both components. By contrast, capacity share responds negatively to idiosyncratic volatility but not to firm volatility.*

Finally, in Table 5, we decompose product demand into a firm-year component and an idiosyncratic product component. We then regress product utilization and capacity share on the absolute value of each component, absorbing firm and product-year fixed effects. The firm fixed effects imply that the estimates are identified from differences across products within the same firm, controlling for time-invariant characteristics.

Both components are negatively related to product-level utilization. Larger common firm-level shocks and larger idiosyncratic product-level shocks are associated with lower utilization, indicating that firms operate product lines less intensively when demand is more volatile or more dispersed around its expected level.

Interestingly, the capacity-share regression reveals a sharper distinction. Capacity shares do not change with the absolute firm-level component but decrease with the idiosyncratic product component. This pattern suggests that common demand variation is associated

with expansion in the firm’s overall capacity allocation, whereas idiosyncratic product risk is associated with a smaller capacity share for the affected product. The distinction is economically useful because it separates shocks that move the firm’s product portfolio jointly from shocks that alter the relative attractiveness of individual product lines within the firm.

TABLE 5: Firm vs. Idiosyncratic Demand Shocks

	Utilization (1)	Capacity Share (2)
Firm shock ($ s_f $)	-0.0424*** (0.0069)	0.0073 (0.0105)
Idiosyncratic shock ($ s_\varepsilon $)	-0.0605*** (0.0090)	-0.0697*** (0.0144)
R^2	0.79	0.79
Firm fixed effects	Yes	Yes
Product $\times$ year fixed effects	Yes	Yes

Estimating equation:

$$y_{ijt} = \beta_f |s_{it}^f| + \beta_\varepsilon |s_{ijt}^\varepsilon| + \alpha_i + \delta_{jt} + \varepsilon_{ijt}$$

where y_{ijt} is either utilization or capacity share, s_{it}^f is the firm-level demand shock (mean of demeaned log sales across products within firm-year), and s_{ijt}^ε is the idiosyncratic product shock (deviation from firm-year mean). We use absolute values to capture volatility/magnitude of shocks. α_i are firm fixed effects and δ_{jt} are product-by-year fixed effects. The time period spans the fiscal years between 1992 to 2016. In our analysis, “year” refers to the fiscal year as opposed to the calendar year because the financial year in India runs from April 1 to March 31. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Taking Stock.

4 Theoretical Framework

We develop a dynamic model of multi-product firms that chooses investment, capacity allocation across product lines, and utilization in response to stochastic product-level demand. The firm accumulates capital over time subject to a one-period time-to-build and a convex adjustment cost. The level of installed capital pins down the total capacity of the firm, which is allocated across products according to expected demand. The remainder of this section sets up the environment and the timing, outlines the dynamic problem, and then solves the

within-period optimization. We use this setup to rationalize the empirical facts and then perform various counterfactuals to underscore [the mechanism] in Section ??.

4.1 Environment

A representative consumer has CES preferences over a continuum of products indexed by ς :

$$C_t = \left(\int D_t(\varsigma)^{1/\eta} c_t(\varsigma)^{(\eta-1)/\eta} d\varsigma \right)^{\eta/(\eta-1)}, \quad \eta > 1, \quad (7)$$

where η is the elasticity of substitution across products and $D_t(\varsigma)$ is a stochastic demand shifter. Utility maximization implies

$$y_t(\varsigma) = D_t(\varsigma) p_t(\varsigma)^{-\eta} C_t P_t^\eta, \quad (8)$$

with P_t the ideal CES price index. We normalize $C_t P_t^\eta = 1$ so that product-level demand at price $p_t(\varsigma)$ is $D_t(\varsigma) p_t(\varsigma)^{-\eta}$. There are $\mathcal{J}$ firms, indexed by j . Each firm produces $\mathcal{I}$ distinct products indexed by $i \in \{1, \dots, \mathcal{I}\}$. The production function for product i at firm j is

$$y_{ijt} = u_{ijt} K_{jt} \omega_{ijt}, \quad (9)$$

where K_{jt} is the firm's installed capacity, $\omega_{ijt} \in [0, 1]$ is the share of capacity allocated to product i (with $\sum_i \omega_{ijt} = 1$), and $u_{ijt} \in [0, 1]$ is the utilization rate of product i 's capacity. The interior assumption $u_{ijt} < 1$ holds under our calibration; we maintain it throughout the analytical results and verify it numerically. Revenue follows from the CES demand structure (8):

$$R_{ijt} = D_{ijt}^{1/\eta} (u_{ijt} K_{jt} \omega_{ijt})^{1-1/\eta}. \quad (10)$$

Utilization is costly:

$$C_{ijt} = \beta K_{jt} \omega_{ijt} u_{ijt}^\gamma, \quad \beta > 0, \gamma > 1. \quad (11)$$

The convexity parameter $\gamma > 1$ captures the increasing marginal cost of pushing utilization toward capacity, reflecting overtime wages (Basu and Kimball, 1996), accelerated depreciation (Greenwood et al., 1988), and other margins through which higher utilization becomes increasingly costly. Operating profit on product i is

$$\Pi_{ijt} = R_{ijt} - C_{ijt} = D_{ijt}^{1/\eta} (u_{ijt} K_{jt} \omega_{ijt})^{1-1/\eta} - \beta K_{jt} \omega_{ijt} u_{ijt}^\gamma. \quad (12)$$

The demand for product i at firm j decomposes into a firm-wide component, a product-

specific mean, and a product-idiosyncratic component:

$$\ln D_{ijt} = f_{jt} + \mu_{ij} + \nu_{ijt}. \quad (13)$$

The firm shock f_{jt} is common to all of firm j 's product lines in year t ; μ_{ij} is a product-specific mean capturing time-invariant differences in average demand across product lines; and ν_{ijt} is a product-firm-year idiosyncratic shock. The two stochastic components follow independent AR(1) processes:

$$f_{jt} = \rho_f f_{j,t-1} + \sigma_f \xi_{jt}, \quad \xi_{jt} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (14)$$

$$\nu_{ijt} = \rho_{\varepsilon,i} \nu_{ij,t-1} + \sigma_{\varepsilon,i} \varepsilon_{ijt}, \quad \varepsilon_{ijt} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (15)$$

with $\xi_{jt} \perp \varepsilon_{kjt}$ for all k, t and $\varepsilon_{ijt} \perp \varepsilon_{kjt}$ for $i \neq k$. The persistence parameters satisfy $\rho_f, \rho_{\varepsilon,i} \in (-1, 1)$. The unconditional variances are

$$s_f^2 \equiv \frac{\sigma_f^2}{1 - \rho_f^2}, \quad s_{\varepsilon,i}^2 \equiv \frac{\sigma_{\varepsilon,i}^2}{1 - \rho_{\varepsilon,i}^2}, \quad s_{D,i}^2 \equiv s_f^2 + s_{\varepsilon,i}^2, \quad (16)$$

where $s_{D,i}^2$ is the total unconditional log-demand variance for product i . The decomposition (13) is the structural counterpart to a one-factor model: a firm-wide common factor that moves all of a firm's products together, plus product-specific noise. Capital accumulation

$$K_{jt+1} = (1 - \delta)K_{jt} + i_{jt}, \quad \delta \in (0, 1), \quad (17)$$

and quadratic adjustment costs $\Phi(i_{jt}, K_{jt}) = \frac{\theta}{2}(i_{jt}/K_{jt})^2 K_{jt}$, $\theta > 0$.

4.2 Timing

Capital and capacity shares are committed before demand is realized, while utilization is chosen after: ex-ante choices made under uncertainty must balance the volatility of one product line against another, while the ex-post utilization decision absorbs whatever demand realization actually occurs. We summarize the timing and information revelation in Figure 1. Let

$$s_{jt} \equiv (f_{jt}, \nu_{1jt}, \dots, \nu_{\mathcal{I}jt}) \quad (18)$$

denote the firm's state at the end of period t —the realized values of the firm-common shock and the product-idiosyncratic shocks. Within period t , decisions and information unfold in the following order. The firm enters period t with predetermined capital K_{jt} and a

predetermined capacity allocation $\{\omega_{ijt}\}_{i=1}^I$. The capacity allocation across product lines is set by the rule $\omega_{i,t} = \Omega_i(s_{t-1})$. The shocks ξ_{jt} and $\{\varepsilon_{ijt}\}_{i=1}^I$ then realize; the firm observes the period- t state s_{jt} and, through (13), the realized product-level demands D_{ijt} . Given $\{D_{ijt}\}_{i=1}^I$ and the predetermined $(K_{jt}, \{\omega_{ijt}\}_{i=1}^I)$, the firm chooses utilization u_{ijt} for each product line, generating output, revenues, and operating profits. At the end of period t , the firm chooses investment i_{jt} . Dividends are then paid,

$$d_{jt} = \pi_{jt} - i_{jt} - \Phi(i_{jt}, K_{jt}), \quad (19)$$

where $\pi_{jt} = \sum_i \Pi_{ijt}$ is total operating profit.

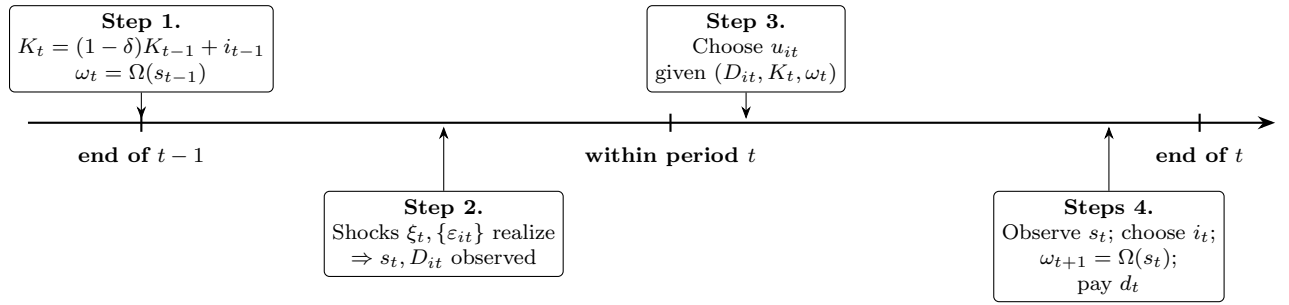


FIGURE 1: Timing in the model.

4.3 Firm Problem

Utilization. After demand is realized in Step 3 in Figure 1, the firm chooses utilization u_{ijt} given $(D_{ijt}, K_{jt}, \omega_{ijt})$. This is a static product-by-product problem. Given ω_{ijt} and realized D_{ijt} , the optimal utilization is

$$u_{ijt}^*(D_{ijt}, \omega_{it}) = \left(\frac{\eta - 1}{\gamma \beta \eta} \right)^{\eta \psi} \left(\frac{D_{ijt}}{K_{jt} \omega_{ijt}} \right)^{\psi}, \quad (20)$$

where $\psi \equiv \frac{1}{\eta(\gamma-1)+1}$ is the elasticity of utilization to demand shocks. Two distinct economic forces pin down ψ . The *cost channel* operates through γ : more convex utilization costs make it expensive to push capacity hard, so the firm smooths its response to demand realizations and ψ falls. The *markup channel* operates through η : with more elastic demand (lower markup), the firm has weaker incentive to push output because additional sales come at a sharper price decline, again lowering ψ .

Firm value function. The value of firm is

$$V_t(K_{jt}, \{\omega_{ijt}\}_{i=1}^{\mathcal{I}}, s_{jt}) = \pi_t^*(K_{jt}, \{\omega_{ijt}\}_{i=1}^{\mathcal{I}}, s_{jt}) + \max_{K_{jt+1}, \{\omega_{ijt+1}\}_{i=1}^{\mathcal{I}}} \left\{ -i_t - \Phi(i_{jt}, K_{jt}) + \frac{1}{1+r} \mathbb{E}_t [V_{t+1}(K_{jt+1}, \{\omega_{ijt+1}\}_{i=1}^{\mathcal{I}}, s_{jt+1})] \right\} \quad (21)$$

Thus, the firm chooses K_{t+1} after observing the current state s_t , while the capacity plan ω_{t+1} is fixed before D_{t+1} is realized and utilization is chosen only after that realization. Marginal q is defined as:

$$q_t \equiv \frac{\partial V_t(K_{jt}, \{\omega_{ijt}\}_{i=1}^{\mathcal{I}}, s_{jt})}{\partial K_t}.$$

Capacity allocation. At the end of period $t-1$, the firm chooses the allocation ω_{ijt} to be deployed in period t . This choice conditions on the observed state s_{t-1} and the anticipated within-period utilization response.

Proposition 1. *The capacity choice is*

$$\omega_{ijt} = \Omega_i(s_{jt-1}) \equiv \frac{M_{ijt|t-1}^{1/\phi}}{\sum_k M_{kjt|t-1}^{1/\phi}}, \quad M_{ijt|t-1} \equiv \mathbb{E}_{t-1}[D_{ijt}^\phi], \quad (22)$$

where $\phi = \gamma\psi$ is the elasticity of operating profit to product-level demand.

Proof. See Appendix C.1.

The optimal capacity share has a transparent economic structure. In our environment, $\phi \equiv \gamma\psi < 1$ for $\gamma, \eta > 1$. Recall $M_i = \mathbb{E}[D_i^\phi]$, two forces shape the allocation Ω_i . The *level channel* tilts capacity toward products with higher expected demand: under lognormal D_i with $\mathbb{E}[D_i] = \bar{D}_i$ held fixed, $M_i^{1/\phi} \propto \bar{D}_i \cdot \exp(-\frac{1}{2}(1-\phi)s_i^2)$ (See Appendix C.1 for the derivation), so the leading term is proportional to mean demand. The *Jensen Inequality channel* works through the exponent on s_i^2 : because profit is concave in demand, a mean-preserving spread reduces expected profit, and the firm tilts capacity away from volatile product lines. The strength of this penalty is governed by $(1-\phi)$: more concave profit (larger $1-\phi$, i.e. lower demand elasticity η or more convex utilization costs γ) produces a sharper allocation response to volatility. Notably, because M_i depends only on the marginal distribution of D_i , the cross-product correlation structure of demand does not enter ω_i^* : two firms with identical marginal distributions but very different cross-product correlations face identical allocation problems.

Further, the allocation of capacity does not depend on the firm size K , and it is fully pinned down by the expected product-level demand that is provided by the information set

s_{jt-1} . Thus, this reduces the dimensionality of the firm problem such that the firm state space collapses to $\{K_{jt}, s_{jt}\}$.

Steady state comparison. Here, we examine how product-level utilization rates are related to demand volatility in steady states. We uncover the structural parameters that derive this relationship.

Proposition 2 (Expected utilization). *Under the shock structure $\ln D_{it} = f_t + \mu_i + \nu_{it}$ with unconditional variances s_f^2 and $s_{\varepsilon,i}^2$, expected steady-state utilization is*

$$\mathbb{E}[u_i^{*,ss}] = \tilde{\Lambda} \cdot \exp\left(-\frac{1}{2}(\gamma - 1)\psi^2(s_f^2 + s_{\varepsilon,i}^2)\right), \quad (23)$$

where $\tilde{\Lambda} \equiv A \cdot [\alpha_1(1 - \phi)]^{-1/\gamma} [(1 + \theta\delta)(r + \delta) - \frac{\theta}{2}\delta^2]^{1/\gamma}$ is a constant.

Proof. See Appendix ??.

In the stationary benchmark, expected utilization is *independent of μ_i* : at ω^* , the capacity-reallocation channel exactly offsets the direct effect of mean demand on utilization. Expected utilization declines with *both* firm-level volatility s_f and product-idiosyncratic volatility $s_{\varepsilon,i}$, entering symmetrically through the total variance $s_f^2 + s_{\varepsilon,i}^2$ with coefficient $(\gamma - 1)\psi^2 > 0$.

The factor $(\gamma - 1)$ measures the convexity of utilization costs. The factor ψ^2 measures utilization variance per unit of demand variance, since at the joint optimum $\ln u_i^*$ is linear in $\ln D_i$ with slope ψ . The two factors move in *opposite directions* as γ changes: more convex costs raise $(\gamma - 1)$ but lower ψ , because a steeper cost wall forces the firm to smooth its utilization response. The Jensen drag is the product of these two opposing forces, and it vanishes at both ends. With linear costs ($\gamma \rightarrow 1$), the firm allows large utilization swings but pays no convex-cost penalty; with extremely convex costs ($\gamma \rightarrow \infty$), the firm essentially freezes utilization, so there is no variance to penalize. The drag is positive only at interior γ where utilization is meaningfully variable *and* cost is meaningfully convex, and it is maximized at $\gamma^* = 1 + 1/\eta$ (close to our calibrated $\hat{\gamma} = 1.3$ for $\eta = 5$). The parameter η shapes the drag through ψ alone: more elastic demand (higher η) weakens the firm's incentive to push utilization with demand, lowers ψ , and so shrinks the Jensen drag.

Lemma 1 (Steady-state capacity allocation). *In the steady-state benchmark with lognormal demand and unconditional variances $s_f^2, s_{\varepsilon,i}^2$, the optimal capacity share satisfies*

$$\omega_i^* \propto \bar{D}_i \cdot \exp\left(-\frac{1}{2}(1 - \phi)s_{\varepsilon,i}^2\right).$$

Two consequences follow from this Lemma. The firm-wide variance s_f^2 does not affect capacity allocation: a common shock that scales every product’s demand by the same factor creates no incentive to shift capacity across lines. Second, holding mean demand fixed, the share is strictly decreasing in the product-idiosyncratic variance with elasticity $-(1-\phi)/2 < 0$. More volatile lines receive less capacity; the strength of the tilt is governed by $(1-\phi)$, the concavity of profit in demand shock.

Lemma 2 (Joint optimum vs. fixed allocation). *The coefficient on $s_{D,i}^2$ that determines how strongly demand volatility lowers expected utilization differs between two benchmarks. At fixed Ω (not optimized), a mean-preserving-spread calculation gives a coefficient of $\eta(\gamma-1)\psi^2$. At the joint optimum Ω^* , the coefficient is $(\gamma-1)\psi^2$ —smaller by a factor of η . The reduction comes from the reallocation channel: volatile products receive less capacity, and because $u^* \propto \omega^{-\psi}$ a smaller share mechanically raises utilization on those lines, so the volatility-induced drop in expected utilization is smaller than it would be without reallocation.*

Capacity reallocation is a powerful internal mechanism through which multi-product firms manage demand risk. By tilting capacity toward less volatile product lines, the firm dampens the “the volatility-induced drop in expected utilization” that demand volatility would otherwise impose on expected utilization, profits, and firm value. The next section quantifies how much of cross-firm differences in size and profitability this reallocation channel explains.

4.4 Estimation and Calibration

We estimate the model on the Prowess product-firm-year panel by the method of simulated moments. We implement the model with two product lines and map them in the data to each firm’s two largest products by total sales over the sample period (product 1 the larger line, without loss of generality) using Prowess’s product classification, whose product codes are consistent across years, so we follow each product line over time; the descriptive facts use all of a firm’s products, whereas the structural estimation uses the leading two to match the two-product line model. On average these two products account for 96% of a firm’s sales, so the two-product specification captures the bulk of its activity.

Our calibration strategy consists of two parts. We first fix a subset of parameters externally based on standard values and direct data counterparts. We then take these external parameters as given and use a simulated method of moments (SMM) procedure to estimate the remaining parameters by matching a targeted set of moments. We describe each step in detail.

Externally set Parameters. We fix the demand elasticity fixed at $\eta = 5$ (a standard value implying a steady-state markup $\eta/(\eta - 1) = 1.25$). The depreciation rate $\delta = 0.08$ is set to the average depreciation rate in the data, and the adjustment cost $\theta = 1$ and interest rate $r = 0.064$ to standard values.

Simulated Method of Moments. We choose the utilization-technology parameters (γ, β) , the firm-specific and product-specific AR(1) innovation variance and persistence parameters $(\sigma_f^2, \rho_f; \sigma_{\varepsilon,i}^2, \rho_{\varepsilon,i})$, and the product demand-mean gap $\mu_1 - \mu_2$ to minimize the weighted distance between a vector of model-simulated moments and their data counterparts. The targeted moments are computed by the same code on the simulated and the Prowess panels: the mean and within-firm cross-product dispersion of capacity utilization; the variances, cross-product covariance, and first-order autocorrelations of demeaned log product output; and the average sales share of the leading product.⁴ Each parameter is disciplined primarily by one moment: the cost scale β by the mean utilization level; the curvature γ by the volatility and within-firm dispersion of utilization; the firm-common variance σ_f^2 by the cross-product covariance of output; the idiosyncratic variances $\sigma_{\varepsilon,i}^2$ by the own output variances; the persistences by the corresponding autocorrelations; and the demand-mean gap by the sales share.

Table 6 reports the estimates and Table 7 the moment fit, which is close across all ten targets. The estimated demand is highly volatile (unconditional idiosyncratic volatilities $s_{\varepsilon,i} = \{1.32, 1.61\}$ and firm-common $s_f = 1.34$), reflecting the high volatility of product-level output and utilization in the data; product 1 is the larger, more persistent ($\rho_{\varepsilon,1} = 0.62$), and somewhat less volatile line.

Table 7 reports the moments of the calibrated baseline. The firm operates well below installed capacity, with mean firm-level utilization of 0.53, and runs its two lines at materially different rates, the within-firm utilization dispersion in Table 7 being the model counterpart of Fact 1’s within-firm spread. Allocation concentrates on the larger line: product 1 takes a sales share of 0.78 versus 0.22 for product 2, matching the data; the tilt away from the more volatile line highlights the underlying capacity allocation, i.e, the firms allocate capacity away from highly volatile product lines. The mean investment rate is $0.08 \approx \delta$ – reflecting the underlying depreciation rate –, and average Tobin’s q is 4.3. Operating profit is strongly right-skewed (mean 0.51 vs. median 0.29), reflecting lognormal demand.

⁴Because physical product output has extreme, lumpy tails, the log-output moments use observations winsorized at the 2.5% level

TABLE 6: Calibrated model parameters.

Parameter	Symbol	Value	Source / Discipline
Demand elasticity	η	5.000	Set externally
Utilization curvature	γ	1.165	Estimated (SMM)
Utilization cost scale	β	1.057	Estimated (SMM)
Depreciation rate	δ	0.080	Matched to data
Interest rate	r	0.064	Set externally
Adjustment cost parameter	θ	1.000	Set externally
Firm shock persistence	ρ_f	0.445	Estimated (SMM)
Firm shock innovation variance	σ_f^2	1.436	Estimated (SMM)
Product demand log-mean	$\{\mu_1, \mu_2\}$	$\{0.926, -0.926\}$	Estimated (SMM)
Product idiosyncratic persistence	$\{\rho_{\varepsilon,1}, \rho_{\varepsilon,2}\}$	$\{0.623, 0.299\}$	Estimated (SMM)
Product idiosyncratic innovation variance	$\{\sigma_{\varepsilon,1}^2, \sigma_{\varepsilon,2}^2\}$	$\{1.068, 2.356\}$	Estimated (SMM)
Product idiosyncratic volatility (uncond.)	$\{s_{\varepsilon,1}, s_{\varepsilon,2}\}$	$\{1.321, 1.609\}$	Derived

4.4.1 Untargeted Moments

How utilization and the capacity allocation respond to demand volatility is not targeted in estimation, so these responses provide an out-of-sample test of the model; we set them against empirical Facts 2 and 4. We run two controlled experiments. In each of these experiments, we hold a product line’s mean demand fixed, so that only the volatility of its demand changes, and we trace the line’s mean utilization and its capacity share ω_i . The first experiment raises the line’s *idiosyncratic* volatility s_ε , holding the firm-common component fixed; the second experiment raises the *firm-common* volatility s_f shared by both lines. Table 8 reports the slopes and Figure 2 the curves.

The model reproduces two features of the data and misses a third. First, utilization falls with volatility from either source (Fact 2): the firm runs more volatile lines with more slack, a precautionary response to demand risk. Second, the capacity allocation responds to *idiosyncratic* volatility but is nearly *invariant* to firm-common volatility. The model slope of the capacity share on idiosyncratic volatility is -0.13 , close to the data coefficient of -0.12 in Fact 4, whereas its slope on firm-common volatility is essentially zero. This follows directly from the optimal rule ω_i (Lemma 1): the firm-specific variance enters both lines identically and cancels from the within-firm shares, leaving only idiosyncratic volatility to

TABLE 7: Calibration moments. The targeted block reports the ten auxiliary moments matched by SMM, computed by the same code on the Prowess product-firm-year panel and on the simulated model; the output (co)variance and autocorrelation moments use demeaned log output winsorized at 2.5%. The additional block reports model-implied quantities that are not targeted in estimation.

Moment	Data	Model
<i>Targeted moments</i>		
Mean utilization	0.601	0.516
Std. dev. of log utilization	0.995	0.879
Within-firm utilization dispersion	0.323	0.254
Mean sales share (leading product)	0.776	0.777
Variance of log output, product 1	0.857	0.851
Variance of log output, product 2	0.959	1.047
Cross-product covariance of log output	0.187	0.193
Autocorrelation of log output, product 1	0.558	0.619
Autocorrelation of log output, product 2	0.470	0.480
Autocorrelation of firm-mean log output	0.527	0.470
<i>Additional model-implied moments</i>		
Investment rate	–	0.083
Tobin’s q	–	4.321
Operating profit	–	0.509
Capital K	–	1.490

tilt capacity across lines. Because only the idiosyncratic component reallocates capacity, the share is weakly related to *total* demand volatility (Fact 3) even as it responds strongly to the idiosyncratic part (Fact 4). On the third count the model falls short: in the data the capacity share rises slightly with firm-common volatility (a coefficient of +0.05), which the model does not produce.

TABLE 8: Untargeted moments. Slopes of product utilization and of the product capacity share on idiosyncratic and on firm-common demand volatility (the standard deviation of the corresponding log-demand component). Neither is targeted in estimation: each is the slope from a controlled experiment that raises one volatility component while holding the model’s mean demand fixed. The data columns report the within-firm coefficients from Fact 2 (utilization) and Fact 4 (capacity share, decomposed).

Response	Slope on demand volatility	
	Data	Model
Utilization, idiosyncratic vol. s_ε	-0.060	-0.092
Utilization, firm-common vol. s_f	-0.042	-0.119
Capacity share, idiosyncratic vol. s_ε	-0.069	-0.129
Capacity share, firm-common vol. s_f	0.0073	0.000

4.5 Counterfactuals

We run three counterfactuals. Each states a result, describes the model experiment behind it, and reports the figure or table it generates. Unless noted, the baseline is the calibrated model under the state-dependent planning rule $\omega_t = \Omega(s_{t-1})$, and the two-product implementation is used throughout.

4.5.1 The value of within-firm capacity reallocation

Shutting down dynamic capacity reallocation (freezing the capacity plan at the unconditional optimum ω^*) reduces steady-state profits by 3.1% and aggregate capital by 2.0%. The loss rises monotonically with the level of idiosyncratic demand volatility, since reallocation is more valuable the more the firm’s product lines fluctuate independently. Capacity reallocation is, in effect, a form of internal insurance, worth on average about 3% of firm profits.

The baseline lets the firm reset its capacity plan every period to the conditionally optimal rule $\omega_t = \Omega(s_{t-1})$, so shares respond to the lagged product-level states. We compare it to a counterfactual that *freezes* the capacity plan at the unconditional optimum $\omega_i^* = M_i^{1/\phi}/S$, $M_i = \mathbb{E}[D_i^\phi]$ (Proposition ??), which shuts down dynamic reallocation while preserving the

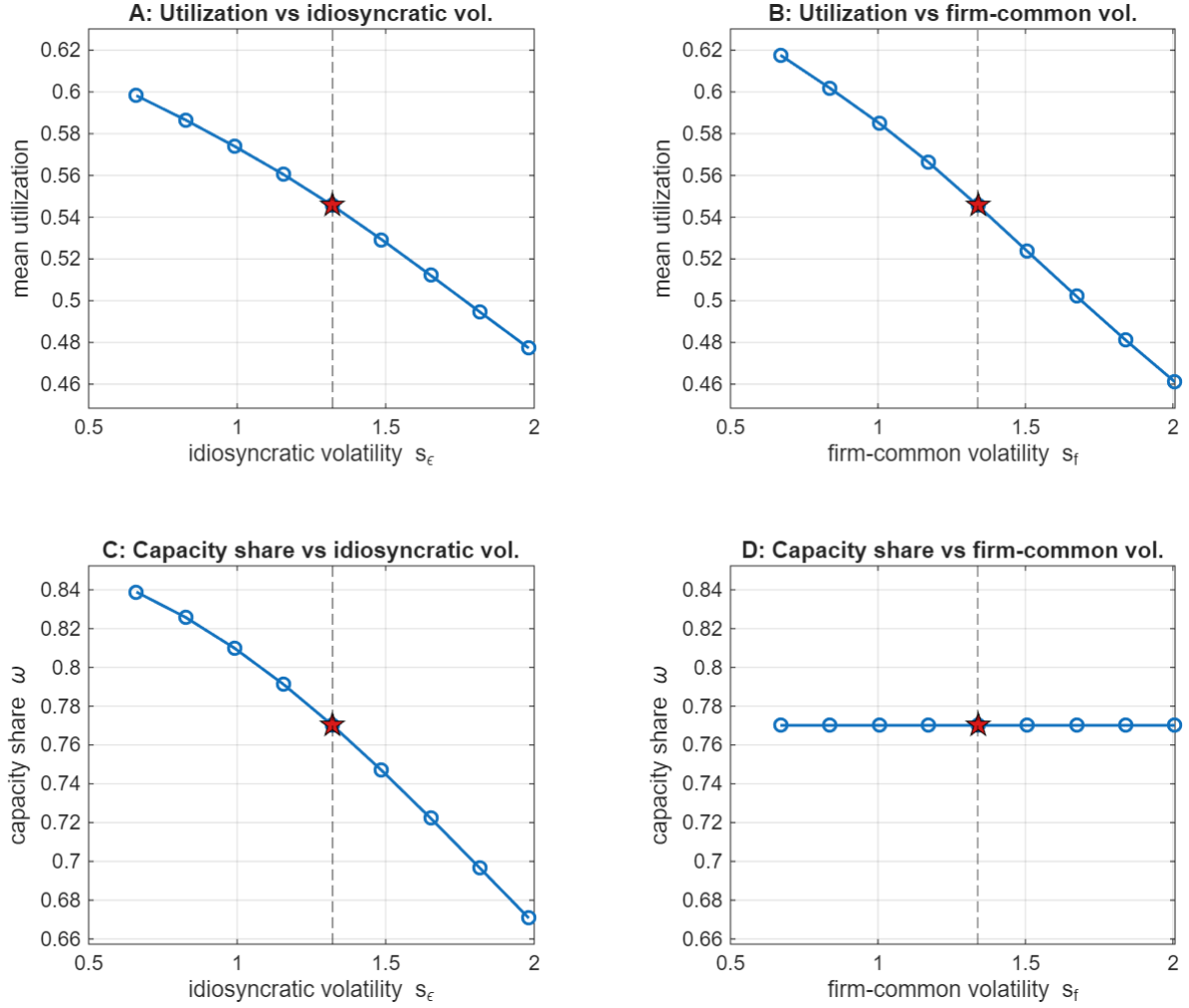


FIGURE 2: Untargeted moments. Mean utilization (top row) and capacity share ω (bottom row) of a product line as its idiosyncratic demand volatility s_ϵ (left column) or the firm-common volatility s_f (right column) rises, holding mean demand fixed. Utilization falls with both volatility components; the capacity share falls with idiosyncratic volatility but is invariant to firm-common volatility. The marker denotes the benchmark calibration.

optimal *average* shares. This isolates the value of dynamic reallocation. To trace the heterogeneity, we start from the calibration and scale both product lines' idiosyncratic volatility up and down together by a common multiplier, holding the common firm volatility s_f and the within-firm balance between the two lines fixed; this varies the *level* of idiosyncratic risk while leaving the common component untouched. Figure 3 reports the steady-state profit and capital loss against the resulting idiosyncratic volatility level s_ε ; the loss rises monotonically, with the calibrated firm (the highlighted marker) in the interior. The value of dynamic reallocation comes from tilting capacity toward whichever line is currently strong, so it grows with the idiosyncratic volatility that makes the two lines diverge; the common firm shock moves both lines together and creates no reallocation motive.

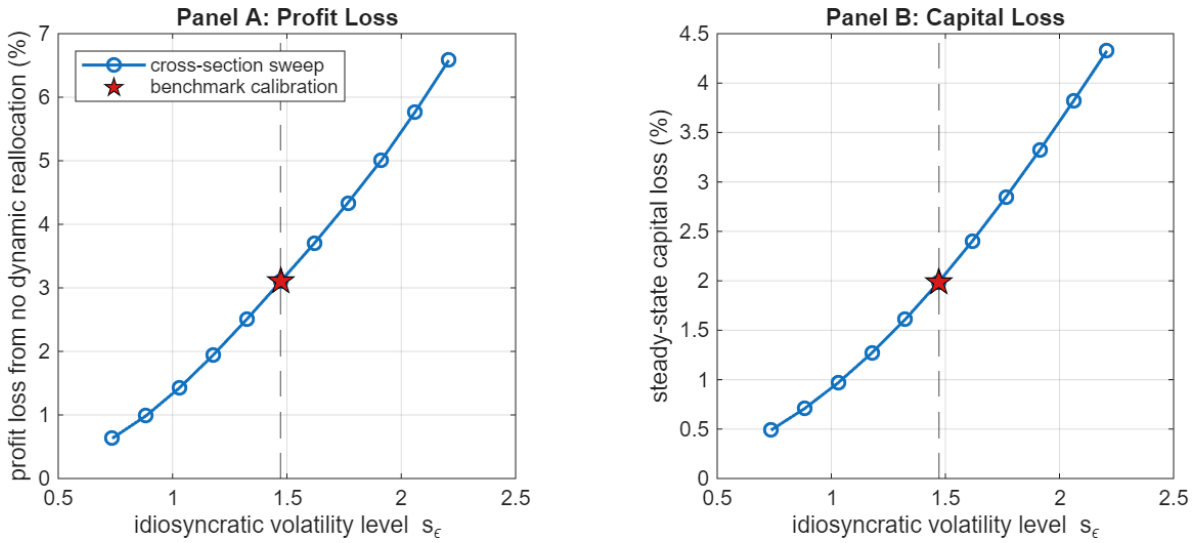


FIGURE 3: Steady-state profit loss (Panel A) and capital loss (Panel B) from freezing the capacity plan at ω^* instead of reallocating, plotted against the level of idiosyncratic demand volatility s_ε (both product lines scaled together, holding the firm shock s_f and the within-firm balance fixed). The dashed line and marker indicate the benchmark calibration.

4.5.2 Same firm-level volatility, different internal economics

Among firms with the same firm-level demand volatility, those whose shocks are predominantly idiosyncratic exhibit $2.4\times$ more within-firm utilization dispersion, $3.1\times$ more active capacity reallocation, and 8.9% lower firm-level profit volatility than otherwise identical firms. Firm-level observables thus hide first-order differences in internal economics that only product-level data can reveal.

We hold each product's marginal demand distribution fixed: the same log-mean μ_i and the same total log-demand variance $s_{D,i}^2 = s_f^2 + s_{\varepsilon,i}^2$. We then move only the split of that variance between the firm-common component s_f^2 , shared by both product lines, and the

product components $s_{\varepsilon,i}^2$, specific to each line. Concretely, we raise the common share s_f/s_D by increasing s_f^2 and lowering each $s_{\varepsilon,i}^2$ by the same amount, so that every product's total variance $s_{D,i}^2$ stays at its calibrated value; we hold the persistences ρ_f and $\rho_{\varepsilon,i}$ fixed and back out the implied innovation variances. Because each product's marginal distribution is unchanged, the firm-level (observable) demand volatility is identical across these firms, and only the cross-product correlation moves, since that correlation is generated entirely by the shared component s_f . At each value of s_f/s_D we simulate the model and record three within-firm objects: the dispersion of utilization across product lines, the capacity-reallocation activity (the time-series standard deviation of the plan share ω_1), and the firm-level profit volatility. Figure 4 contrasts a firm whose demand variance is dominated by the product component (low s_f/s_D) with one dominated by the firm common component (high s_f/s_D), plotting each object against s_f/s_D . Consistent with Proposition ?? and the reallocation remark, firms with a larger product component display more utilization dispersion, reallocate more actively, and, through diversification, have lower firm-level profit volatility.

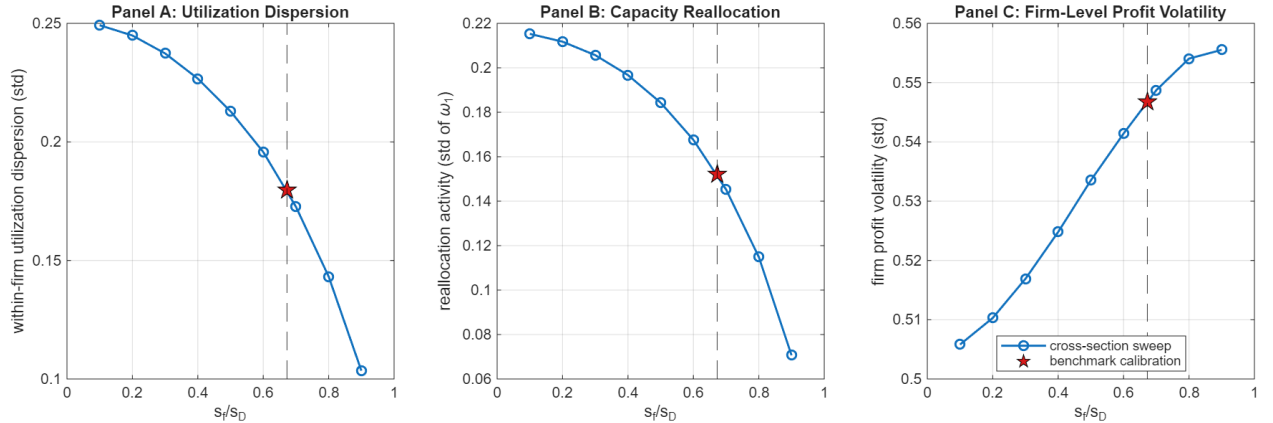


FIGURE 4: Same firm-level volatility, different internal economics. Each panel plots one within-firm object against the firm common share s_f/s_D , holding every product's marginal demand distribution fixed; the marker denotes the benchmark calibration. *Panel A*, utilization dispersion: the cross-product standard deviation of realized utilization within a firm-year, averaged over the simulated panel (with two products this equals $|u_1 - u_2|/\sqrt{2}$); it measures how differently the firm runs its two product lines. *Panel B*, capacity reallocation: the within-firm time-series standard deviation of the planned capacity share ω_1 , averaged across firms; it measures how actively the firm shifts capacity across product lines over time. *Panel C*, firm-level profit volatility: the within-firm time-series standard deviation of total operating profit, averaged across firms.

4.5.3 Average q , marginal q , and excess capacity

Grullon and Ikenberry (2025) show that when firms expect to operate below full capacity, average q is a biased proxy for the unobservable marginal q that drives investment, and

that adding *asset utilization* (the sales-to-capital ratio) to investment regressions controls for expected excess capacity and restores the relationship. Our model is a natural laboratory: firms optimally operate below full capacity, and because the model delivers the *true* marginal q it lets us examine two things firm-level data cannot. We find that firm-level asset utilization is a near-sufficient statistic for investment even with multiple products, so Grullon and Ikenberry’s firm-level correction loses almost nothing to within-firm composition; but because our excess capacity is optimal underutilization rather than a binding constraint, the average–marginal q wedge behaves oppositely to theirs, narrowing as excess capacity rises.

Consider first Grullon and Ikenberry’s correction in the multi-product firm. We run their specification on the simulated panel: a regression of the investment rate I/K on average q , to which we then add asset utilization $AU = \sum_i \text{sales}_i/K$, their proxy for expected excess capacity. Adding AU enters positively and lifts the R^2 from 0.77 to 0.84 (Table 9), exactly the role their model assigns it: asset utilization controls for the excess-capacity component of average q . The step that only product-level data permit is the next one: replacing firm AU with its product-level components $\{\text{sales}_i/K\}$, entered separately, raises the R^2 by a further $\Delta R^2 \approx 0.008$ only. Firm-level asset utilization is therefore a *near-sufficient statistic for investment* even with multiple product lines. The reason is structural: at the optimal utilization each product’s revenue is $R_i \propto D_i^\phi (K\omega_i)^{1-\phi}$, so firm sales $\sum_i R_i \propto K^{1-\phi} \mathcal{D}$ and hence $MPK \propto \sum_i \text{sales}_i/K$: optimal capacity allocation and utilization integrate out the within-firm composition, a multi-product, Hayashi-type foundation for Grullon and Ikenberry’s use of firm-level asset utilization.

Consider next the wedge itself. Although firm AU absorbs the excess-capacity bias in the investment regression, the average–marginal q wedge behaves very differently from Grullon and Ikenberry. As demand volatility rises (Figure 5; Table 10), utilization falls (more excess capacity), but *marginal* q is pinned by the investment first-order condition (flat at $\approx (1+r)(1+\theta\delta)$), while *average* q falls because lower utilization depresses firm value. The wedge therefore *narrows* as excess capacity rises, the opposite of Grullon and Ikenberry. The reason is that our excess capacity is *optimal underutilization* (a smooth interior choice, $u < 1$), not a binding demand constraint with a positive shadow price: the marginal unit of capital is not trapped behind a constraint, so it carries no excess-capacity wedge. This sharpens the distinction from Grullon and Ikenberry: here, low and dispersed utilization is deliberate hedging, not a friction. [Can we test this in the data?]

TABLE 9: CF3: Grullon–Ikenberry’s investment regression in the multi-product model.

Investment I/K is regressed on average q (col. 1), then adding firm asset utilization $AU = \sum_i \text{sales}_i/K$ (col. 2, GI’s proxy for expected excess capacity), then replacing AU with its product-level components $\{\text{sales}_i/K\}$ (col. 3). Adding AU raises R^2 and enters positively, reproducing GI’s excess-capacity control; the product-level components add almost nothing over firm AU , so firm-level asset utilization is a near-sufficient statistic for investment even with multiple products.

	(1) I/K	(2) I/K	(3) I/K
Average q	0.106	0.078	0.079
Asset utilization $AU = \sum_i \text{sales}_i/K$		0.046	
product 1, sales_1/K			0.051
product 2, sales_2/K			0.009
R^2	0.769	0.843	0.851

ΔR^2 from adding firm AU (col. 2 vs. 1): +0.074; from the product-level components over firm AU (col. 3 vs. 2): +0.008. Simulated firm-year panel; coefficients shown (t -statistics uninformative at this sample size).

TABLE 10: CF3: the average–marginal q wedge vs. demand volatility. Marginal q is pinned by the investment first-order condition (flat); average q falls with volatility, so the wedge *narrows* as excess capacity rises — the opposite of Grullon–Ikenberry, because excess capacity here is optimal underutilization, not a binding demand constraint.

Object	Value
Marginal q , low volatility	1.155
Marginal q , high volatility	1.148
Wedge (avg – marginal q), low volatility	3.473
Wedge (avg – marginal q), high volatility	2.180

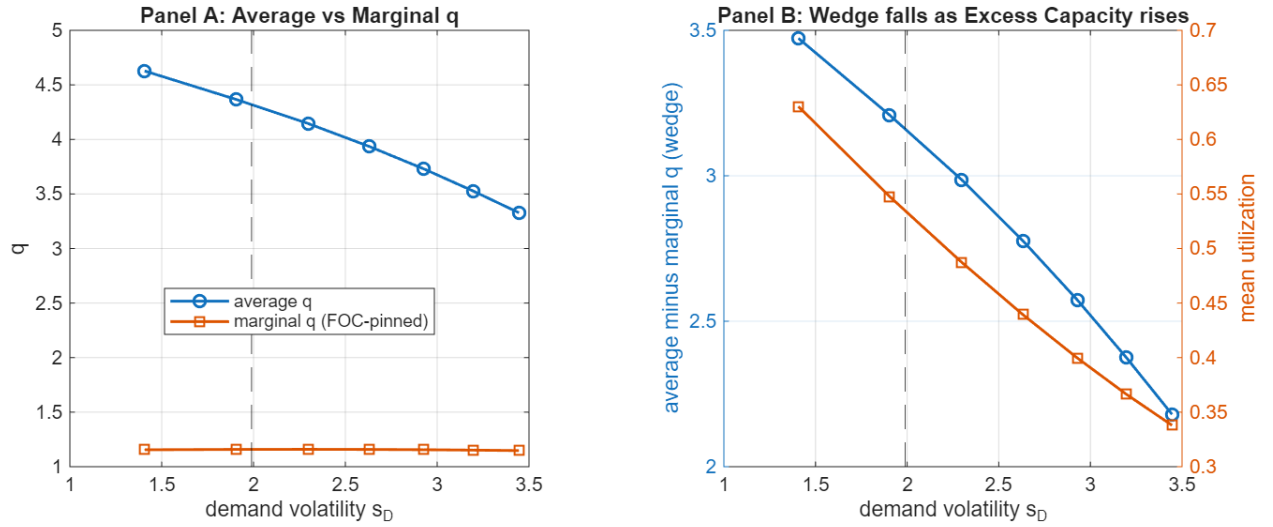


FIGURE 5: The average–marginal q wedge as demand volatility s_D rises (both the firm-common and product-specific components scaled up; each product’s mean held fixed). *Panel A*: average q falls with volatility while marginal q stays flat, pinned by the investment first-order condition. *Panel B*: the wedge (left axis) narrows even as mean utilization (right axis) falls, i.e. as excess capacity rises, the opposite of Grullon and Ikenberry, because here excess capacity is optimal underutilization rather than a binding demand constraint. The dashed line marks the benchmark calibration.

5 Conclusion

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INTERNET APPENDIX FOR ONLINE PUBLICATION

A Additional figures and tables

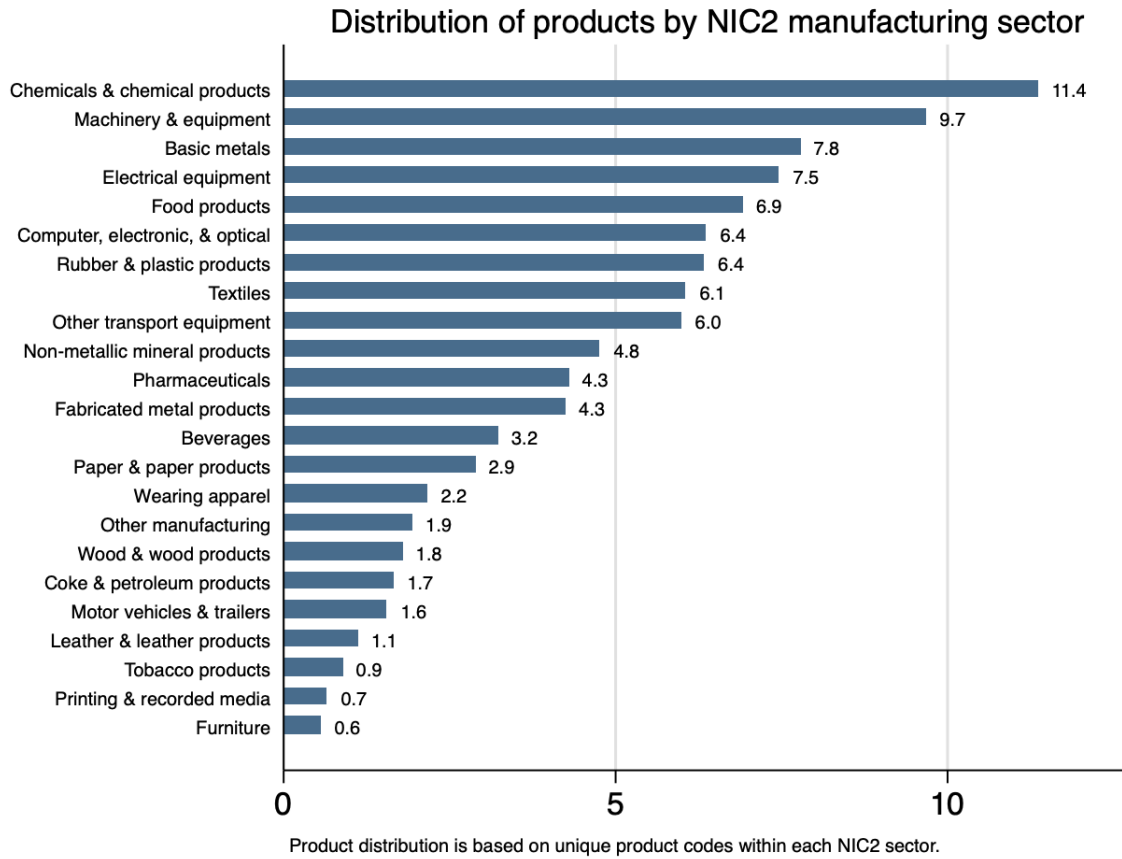


FIGURE IAA1: Industry Distribution: Product Coverage

This figure reports the distribution of products across NIC2 manufacturing sectors in the sample. For each sector, it plots the share of unique product codes assigned to that sector out of all unique products observed in the manufacturing sample. This figure shows how broadly product coverage is distributed across industries and helps document whether the product-level data are concentrated in a few sectors or span a wide cross-section of manufacturing activity.

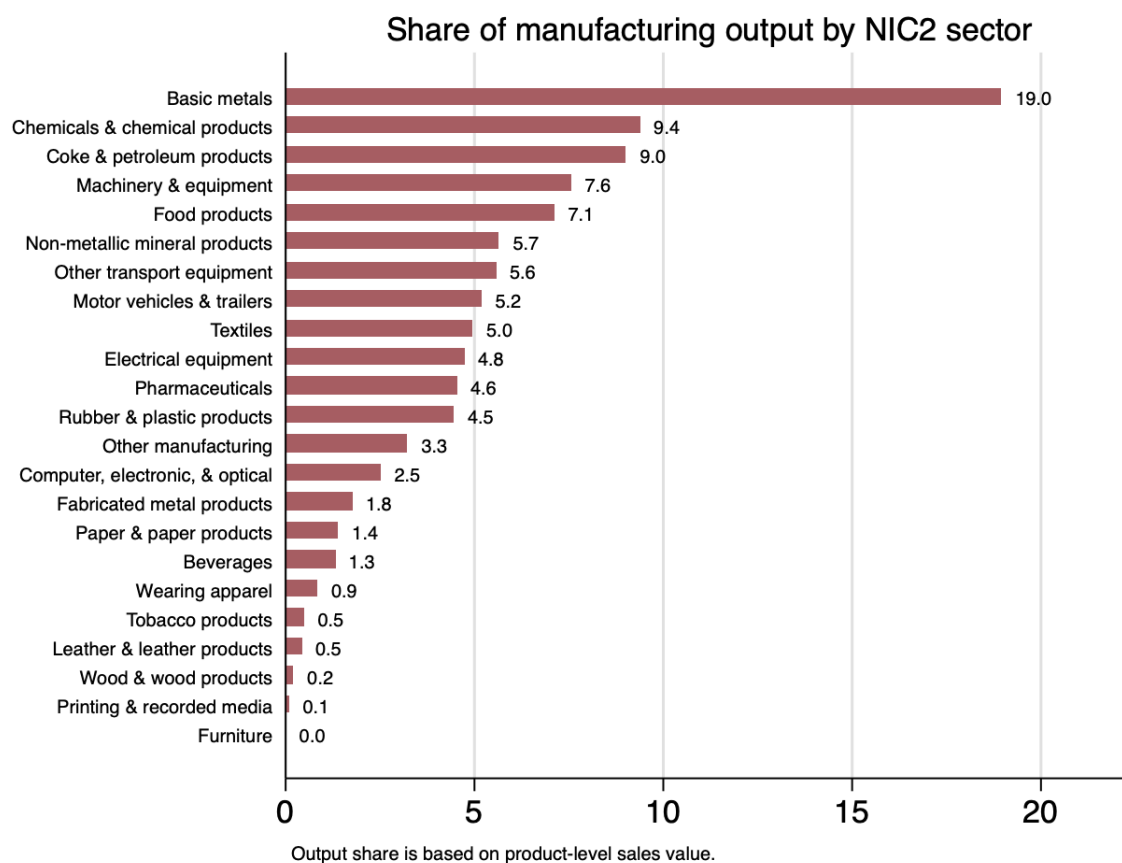


FIGURE IAA2: Industry Distribution: Output Share

This figure reports each NIC2 manufacturing sector's share of total manufacturing output in the sample. Output is measured using product-level sales value, aggregated within each NIC2 sector and divided by total sales value across all manufacturing sectors. This figure complements the product-count distribution by showing the economic weight of each sector, rather than simply the number of products observed.

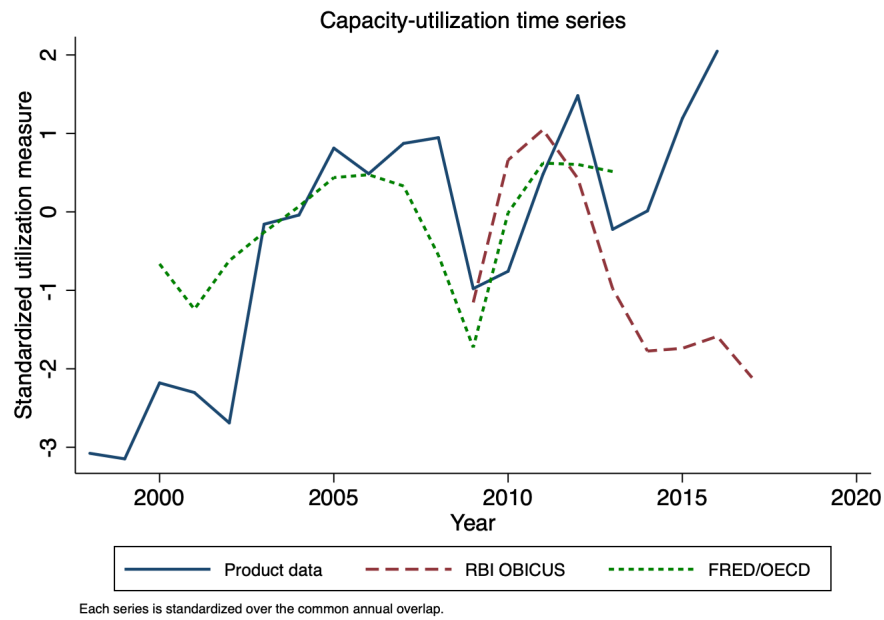


FIGURE IAA3: Capacity Utilization

This figure reports the distribution of products across NIC2 manufacturing sectors in the sample. For each sector, it plots the share of unique product codes assigned to that sector out of all unique products observed in the manufacturing sample. This figure shows how broadly product coverage is distributed across industries and helps document whether the product-level data are concentrated in a few sectors or span a wide cross-section of manufacturing activity.

B Estimating Quantity-Based Productivity

Estimating productivity is difficult in environments where firms face frictions in adjusting inputs. Capital constraints, unreliable electricity, infrastructure bottlenecks, and labor regulations can all weaken the link between observed input choices and underlying productivity. Standard proxy-variable approaches, such as [Levinsohn and Petrin \(2003\)](#), rely on the idea that firms can adjust intermediate inputs in response to productivity shocks. These methods typically recover revenue-based productivity, TFPR. While TFPR is informative, it combines physical efficiency with other forces that affect revenues, including markups, product quality, and product mix. In our setting, each of these margins may plausibly respond to emissions regulation.

At the same time, revenue-based measures are not necessarily undesirable. If higher-quality products command higher prices, then prices and revenues may capture economically meaningful differences in firms' ability to produce quality, as emphasized by [Atkin, Khandelwal, and Osman \(2019\)](#). For this reason, we follow the methodology of [De Loecker, Goldberg, Khandelwal, and Pavcnik \(2016\)](#), which provides a way to estimate quantity-based productivity while allowing for systematic differences in product quality. The approach is flexible enough to accommodate a broad class of demand systems and does not require imposing a particular degree of pass-through from input prices to output prices. It also yields estimates of markups and marginal costs at the firm-product-year level.

The key idea is to use variation in input prices to proxy for differences in product quality. We model output quality as an increasing function of output prices, product market shares, and product indicators. This is motivated by the fact that producers selling higher-priced goods tend to use higher-priced inputs, a pattern documented, for example, by [Kugler and Verhoogen \(2012\)](#). Following [De Loecker et al. \(2016\)](#), we also assume that quality is correlated across inputs. Intuitively, producing a high-quality good generally requires not only better materials, but also complementary labor and capital. This lets us summarize input-quality differences using a single firm-product-level quality index.⁵

B.1 Assumptions

The estimation procedure relies on a set of simplifying assumptions, following [De Loecker et al. \(2016\)](#). These assumptions allow us to recover quantity productivity, marginal costs, and markups using the available data.

1. Producers of a given product share the same production technology, although their productivity may differ.
2. A firm has the same productivity level across all of the products it manufactures.
3. In the short run, firms adjust output through materials, while labor and capital are fixed or slow-moving.

⁵Because input prices are strongly related to output prices and other measures of product quality, we use output quality to control for input price variation across firms. The maintained assumption is that input quality is positively correlated across factors of production, so that a single product-quality index can capture systematic variation in the prices of materials, labor, and capital used to produce a given product.

4. Firms minimize contemporaneous short-run costs, taking current output and input prices as given.
5. Production function parameters are stable over the sample period.
6. More productive firms are more likely to operate as multi-product producers.

The second assumption is clearly restrictive, but it is commonly used because it makes it possible to estimate markups for firms that produce multiple goods. The third assumption is needed because labor and capital are observed only at the firm-year level, not separately by product. This assumption does not rule out longer-run substitution between capital and labor; it only implies that, within the short-run markup calculation, materials are treated as the adjustable input.

Under the assumptions that materials are the variable input and that firms minimize costs, markups can be written as the ratio of the output elasticity of materials to the materials share of revenue. The cost-minimization assumption also implies that input prices are not affected by the quantity of inputs purchased. This rules out static input-market power, such as monopsony power. As a result, the level of estimated markups may be biased downward. The estimates are therefore most useful for studying changes in markups rather than their absolute level. This distinction is important for our application: if input-market power is either fixed over time or unrelated to the 2009 emissions regulation, then changes in estimated markups remain informative. Finally, the stability of production coefficients is imposed because the data do not support estimating separate production functions for different subperiods.

B.2 Sources of bias and empirical corrections

Two empirical challenges are central for estimating TFPQ in our setting. First, for firms that make multiple products, we do not observe how inputs are allocated across products. Second, product quality is unobserved.

We address the first issue by estimating the production function using only single-product firms. This avoids the need to allocate inputs across products when recovering the production technology. However, single-product status is itself endogenous. Firms choose whether to produce one or several products, and this decision is likely related to productivity. To correct for this selection problem, we use the assumption that product scope increases with productivity. We estimate the probability that a firm remains a single-product producer as a function of its expected productivity and state variables, including product scope, materials, location, and other exogenous characteristics. The assumption that single- and multi-product firms share the same product-specific technology then allows us to apply the production function estimates from single-product firms to multi-product firms.

The second issue is that product quality is not directly observed. If higher-quality firms produce goods that differ systematically from lower-quality firms within the same product category, then quantity-based productivity estimates can be biased downward when quality is omitted.⁶ We handle this by using input prices as a proxy for input quality, and by linking

⁶TFPR partly captures cross-sectional quality differences because prices enter revenues. However, prices

input quality to output quality. Because product-level input use is not observed for multi-product firms, the production function is again estimated on the single-product subsample. We then use the estimated relationship between quality, prices, market shares, and product characteristics to control for input price differences across firms.⁷

The implementation proceeds in four steps:

1. We estimate product-specific production functions using single-product firms. The production technology is specified as translog, and the estimates are used to recover the output elasticity of materials.
2. We correct for selection into single-product production. Specifically, we estimate the productivity cutoff above which firms expand into multiple products, and control for the probability that a firm remains below that cutoff in a given year as a function of predicted productivity and state variables.
3. For multi-product firms, we impute product-level materials shares using the production function estimates from single-product firms. The allocation of materials across products is modeled as a function of firm-product-year output prices, product market shares, product and location fixed effects, and export status.
4. We compute markups and marginal costs at the firm-product-year level. A product's markup is the ratio of the output elasticity of materials to the materials share of revenue. Marginal cost is then obtained by dividing the product price by the estimated markup.

This procedure gives us quality-adjusted estimates of TFPQ, markups, and marginal costs while allowing for flexible differences in input and output quality across firms.

B.3 Validation exercises

We conduct several diagnostic checks to assess whether the estimates line up with basic economic intuition and with patterns documented in prior work. Figure IAB4 plots the relationship between output quantities and our estimated markups and marginal costs after removing product-year fixed effects.

The left panel shows that firms producing larger quantities tend to have higher markups. This is consistent with the idea that high-output firms have lower marginal costs and can sustain larger markups. The right panel relates marginal costs to quantities. Our estimated production elasticities indicate that many firms operate under increasing returns to scale, a feature also documented by De Loecker et al. (2016). Consistent with this pattern, marginal costs decline with the quantity produced.

also reflect markups and demand-side conditions, so TFPR can confound productivity, quality, and market power. The sign and size of the resulting bias depend on the setting.

⁷In the data, input prices increase with product quality. This supports the use of output-quality proxies, including output prices, product market shares, and product indicators, to account for input price differences across producers. This approach is consistent with evidence that firms producing more expensive goods tend to use more expensive inputs, as in Kugler and Verhoogen (2012).

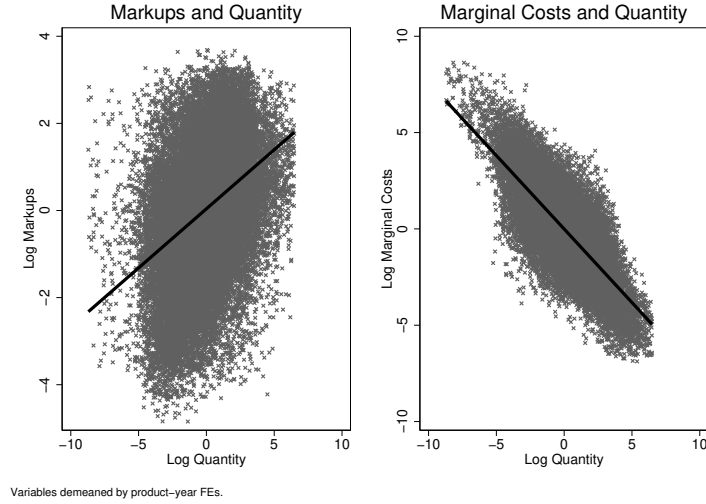


FIGURE IAB4: Marginal Costs, Markups, and Quantities

The left panel plots the relationship between the natural logarithm of product markups and output quantities. The right panel plots the relationship between the natural logarithm of product marginal costs and output quantities. Observations are at the firm-product-year level for 1992–2016. Markups, marginal costs, and quantities are residualized using product-year fixed effects so that comparisons are made across firms within the same product-year.

We also evaluate whether applying production function estimates from single-product firms to multi-product firms produces plausible within-firm patterns. Figure IAB5 examines how markups and marginal costs vary across products within the same firm. To make products comparable within firms, we residualize markups and marginal costs using both product-year and firm-year fixed effects. We then relate these residualized outcomes to each product’s share of the firm’s sales.

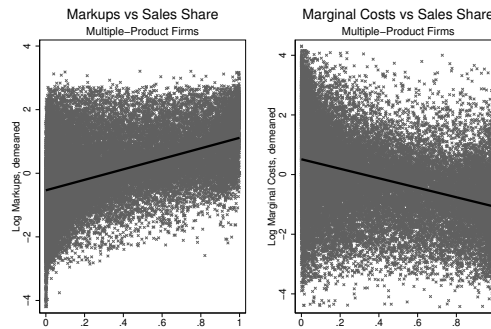


FIGURE IAB5: Markups, Costs and Product Sales Share

Notes: The left panel presents the correlation between the natural log of product markup and sales share. The right panel is between the natural log of product marginal cost and sales share. Data are at the firm-product-year level for the period 2005 to 2015. Data are winsorized at the 3rd and 97th percentiles. Markups, marginal costs, and quantities are demeaned by product-year and firm-year fixed effects to make them comparable across firms.

The patterns are consistent with the idea that multi-product firms have core products.

Products that account for a larger share of firm sales tend to have higher markups and lower marginal costs. Conversely, products that are farther from the firm’s main area of strength have lower markups and higher marginal costs. Although the estimation procedure does not impose a particular demand system or market structure, these correlations are in line with models of multi-product firms, including [Eckel and Neary \(2010\)](#), [Mayer, Melitz, and Ottaviano \(2014\)](#), and [Melitz and Ottaviano \(2008\)](#), as well as with the empirical evidence in [De Loecker et al. \(2016\)](#) for Indian manufacturing.

Production Function Framework. We estimate a gross output production function for each product line following the control function approach ([Levinsohn and Petrin, 2003](#); [Akerberg et al., 2015](#)). The production function takes the form:

$$y_{ijt} = \beta_l l_{ijt} + \beta_k k_{ijt} + \beta_m m_{ijt} + \omega_{ijt} + \varepsilon_{ijt} \quad (\text{B.1})$$

where y_{ijt} is log output, l_{ijt} is log labor, k_{ijt} is log capital, m_{ijt} is log materials, and ω_{ijt} is productivity. We use the proxy variable method to address the simultaneity of input choices and unobserved productivity.

Recovering Material Shares. From the production function estimation, we obtain estimates of the revenue function ρ_{ijt} , which captures the relationship between inputs and revenue. The material cost share for product j in firm i is then computed as:

$$\omega_{ijt}^M = \frac{P_{ijt}Q_{ijt}}{\exp(\hat{\rho}_{ijt})} \quad (\text{B.2})$$

where $P_{ijt}Q_{ijt}$ is product revenue (sales value) and $\hat{\rho}_{ijt}$ is the estimated revenue function from the production function estimation.

Interpretation as Capacity Allocation. The material share ω_{ijt}^M reflects the intensity of variable input usage for each product line. Under the assumption that material inputs scale with production intensity, products receiving a larger share of material inputs represent a greater allocation of the firm’s productive capacity.

We winsorize the material shares at the 5th and 95th percentiles to limit the influence of outliers, yielding our final capacity share measure ω_{ijt} .

Remark. An alternative interpretation relates to the markup estimation literature ([Loecker and Warzynski, 2012](#)). Under cost minimization, the material cost share relates to the output elasticity of materials and the markup:

$$\omega_{ijt}^M = \frac{\theta_{ijt}^M}{\mu_{ijt}} \quad (\text{B.3})$$

where θ_{ijt}^M is the output elasticity of materials and μ_{ijt} is the markup. Our capacity share measure thus also captures variation in production intensity relative to market power across product lines.

C Derivations

C.1 Optimal Utilization (Proposition ??)

First-order condition for $\Pi_{it} = D_{it}^{1/\eta} (u_{it} K_t \omega_{it})^{1-1/\eta} - \beta K_t \omega_{it} u_{it}^\gamma$ with respect to u_{it} :

$$(1 - 1/\eta) D_{it}^{1/\eta} (K_t \omega_{it})^{1-1/\eta} u_{it}^{-1/\eta} = \gamma \beta u_{it}^{\gamma-1} K_t \omega_{it}.$$

Rearranging:

$$u_{it}^{\gamma-1+1/\eta} = \frac{\eta-1}{\gamma\beta\eta} D_{it}^{1/\eta} (K_t \omega_{it})^{-1/\eta}.$$

Note $\gamma - 1 + 1/\eta = [\eta(\gamma - 1) + 1]/\eta = 1/(\eta\psi)$. Raising to $\eta/[\eta(\gamma - 1) + 1] = \eta\psi$:

$$u_{it}^* = \left(\frac{\eta-1}{\gamma\beta\eta} \right)^{\eta\psi} \left(\frac{D_{it}}{K_t \omega_{it}} \right)^\psi = A \left(\frac{D_{it}}{K_t \omega_{it}} \right)^\psi. \quad \blacksquare$$

C.2 Optimal Capacity Allocation (Proposition ??)

Expected profit is separable: $\mathbb{E}[\Pi_{\text{total}}] = \sum_i \mathbb{E}[\Pi_i]$ evaluated at u_i^* . From Stage 2, $(1 - 1/\eta)R_i = \gamma C_i$, so $(1 - 1/\eta)\mathbb{E}[R_i] = \gamma\mathbb{E}[C_i]$.

By the envelope theorem:

$$\frac{\partial \mathbb{E}[\Pi_i]}{\partial \omega_i} = \frac{(\gamma - 1)}{\omega_i} \mathbb{E}[C_i].$$

Equalizing marginal contributions with $\sum_i \omega_i = 1$: $\mathbb{E}[C_i]/\omega_i = \mathbb{E}[C_j]/\omega_j$. Since $\mathbb{E}[C_i] = \beta K \omega_i \mathbb{E}[u_i^{\gamma}]$ and $\mathbb{E}[u_i^{\gamma}] \propto \omega_i^{-\phi} M_i$ (using $\psi\gamma = \phi$), this becomes $M_i/\omega_i^\phi = M_j/\omega_j^\phi$.

Combining with the constraint: $\omega_i^* = M_i^{1/\phi}/S$ where $S = \sum_j M_j^{1/\phi}$. $\blacksquare$

C.3 Profit Representation (Proposition ??)

At the Stage-2 optimum, $(1 - 1/\eta)R_i = \gamma C_i$, so $\Pi_i = R_i - C_i = R_i(1 - (1 - 1/\eta)/\gamma) = R_i \cdot [\eta(\gamma - 1) + 1]/(\eta\gamma)$.

Substituting u^* into $R_i = D_i^{1/\eta} (u_i K \omega_i)^{1-1/\eta}$:

- Exponent on D_i : $1/\eta + \psi(\eta - 1)/\eta = [1 + \psi(\eta - 1)]/\eta = \eta\phi/\eta = \phi$.
- Exponent on $K\omega_i$: $(1 - 1/\eta)(1 - \psi) = (\eta - 1)(\gamma - 1)/[\eta(\gamma - 1) + 1] = 1 - \phi$.

Therefore:

$$\pi_t = \sum_i \Pi_i = \alpha_1 \cdot K_t^{1-\phi} \cdot \sum_i D_{it}^\phi \omega_{it}^{1-\phi} = \alpha_1 K_t^{1-\phi} \mathcal{D}_t. \quad \blacksquare$$

C.4 Steady-State K_{ss} , MPK, and q (Proposition ??)

FOC for K_{t+1} . Differentiating the post-demand Bellman with respect to K_{t+1} in the no-issuance baseline:

$$1 + \theta(i_t/K_t) = \frac{1}{1+r} \mathbb{E}_t[q_{t+1}], \quad q_t \equiv \frac{\partial V_t^{post}}{\partial K_t}.$$

Envelope. Three channels contribute to $\partial V_t^{post}/\partial K_t$: $\partial \pi_t/\partial K_t = MPK_t$; the investment term gives $+(1-\delta)$; the adjustment-cost term gives $-(1-\delta)\theta(i_t/K_t) - \frac{\theta}{2}(i_t/K_t)^2$ (with sign $-$ because we're differentiating $-\Phi$). Summing:

$$q_t = MPK_t + (1-\delta)[1 + \theta(i_t/K_t)] + \frac{\theta}{2}(i_t/K_t)^2.$$

Steady state. At $i/K = \delta$, the FOC gives $q_{ss} = (1+r)(1+\theta\delta)$. The envelope gives $MPK_{ss} = q_{ss} - (1-\delta)(1+\theta\delta) - \frac{\theta}{2}\delta^2 = (1+\theta\delta)[(1+r) - (1-\delta)] - \frac{\theta}{2}\delta^2 = (1+\theta\delta)(r+\delta) - \frac{\theta}{2}\delta^2$.

K_{ss} . Using $MPK_{ss} = \alpha_1(1-\phi)K_{ss}^{-\phi}\mathcal{D}_{ss}$ and inverting:

$$K_{ss} = \left[\frac{\alpha_1(1-\phi)\mathcal{D}_{ss}}{(1+\theta\delta)(r+\delta) - \frac{\theta}{2}\delta^2} \right]^{1/\phi}. \quad \blacksquare$$

C.5 Steady-State Utilization (Proposition ??)

Start from $u_i^*(D_i, \omega_i) = A(D_i/K\omega_i)^\psi$. Plug in $K = K_{ss}$:

$$K_{ss}^{-\psi} = \left[\frac{\alpha_1(1-\phi)\mathcal{D}_{ss}}{(1+\theta\delta)(r+\delta) - \frac{\theta}{2}\delta^2} \right]^{-\psi/\phi} = \left[\frac{\alpha_1(1-\phi)\mathcal{D}_{ss}}{(1+\theta\delta)(r+\delta) - \frac{\theta}{2}\delta^2} \right]^{-1/\gamma}$$

(using $\psi/\phi = 1/\gamma$). Therefore:

$$u_i^{*,ss}(D_i | \omega) = \underbrace{A[\alpha_1(1-\phi)]^{-1/\gamma} [(1+\theta\delta)(r+\delta) - \frac{\theta}{2}\delta^2]^{1/\gamma}}_{\equiv \tilde{\Lambda}} D_i^\psi \omega_i^{-\psi} \mathcal{D}_{ss}^{-1/\gamma}. \quad (\text{C.1})$$

Impose ω_i^* .

(i) *The ω -term.*

$$(\omega_i^*)^{-\psi} = \left(M_i^{1/\phi}/S \right)^{-\psi} = S^\psi M_i^{-\psi/\phi} = S^\psi M_i^{-1/\gamma}.$$

(ii) *The $\mathcal{D}_{ss}$ -term.* At ω^* ,

$$\mathcal{D}_{ss} = \sum_j M_j (\omega_j^*)^{1-\phi} = \sum_j M_j \left(M_j^{1/\phi}/S \right)^{1-\phi} = S^{-(1-\phi)} \sum_j M_j^{1+(1-\phi)/\phi} = S^{-(1-\phi)} \sum_j M_j^{1/\phi} = S^\phi,$$

using $1 + (1 - \phi)/\phi = 1/\phi$. Raising to $-1/\gamma$ and using $\phi/\gamma = \psi$:

$$\mathcal{D}_{ss}^{-1/\gamma} = S^{-\phi/\gamma} = S^{-\psi}.$$

(iii) *Combine.*

$$(\omega_i^*)^{-\psi} \mathcal{D}_{ss}^{-1/\gamma} = \left(S^\psi M_i^{-1/\gamma} \right) (S^{-\psi}) = M_i^{-1/\gamma}.$$

Plugging into (C.1):

$$u_i^{*,ss}(D_i) = \tilde{\Lambda} D_i^\psi M_i^{-1/\gamma}. \quad \blacksquare$$

C.6 Expected Utilization under Lognormal Demand (Proposition ??)

Under the shock structure $\ln D_{it} = f_t + \mu_i + \nu_{it}$ with f_t, ν_{it} independent normal processes, the unconditional distribution of $\ln D_{it}$ is Gaussian with mean μ_i and variance $s_{D,i}^2 \equiv s_f^2 + s_{\varepsilon,i}^2$. Therefore:

$$\mathbb{E}[D_i^\psi] = \exp\left(\psi\mu_i + \frac{1}{2}\psi^2 s_{D,i}^2\right), \quad M_i = \mathbb{E}[D_i^\phi] = \exp\left(\phi\mu_i + \frac{1}{2}\phi^2 s_{D,i}^2\right).$$

Therefore:

$$M_i^{-1/\gamma} \mathbb{E}[D_i^\psi] = \exp\left[\left(-\frac{\phi}{\gamma} + \psi\right)\mu_i + \frac{1}{2}\left(-\frac{\phi^2}{\gamma} + \psi^2\right)s_{D,i}^2\right].$$

Coefficient on μ_i : $-\phi/\gamma + \psi = -\psi + \psi = 0$ (using $\phi = \gamma\psi$). So μ_i drops out.

Coefficient on $s_{D,i}^2$: $-\phi^2/\gamma + \psi^2 = -\gamma\psi^2 + \psi^2 = -(\gamma - 1)\psi^2$.

Therefore:

$$\mathbb{E}[u_i^{*,ss}] = \tilde{\Lambda} \cdot \exp\left(-\frac{1}{2}(\gamma - 1)\psi^2 (s_f^2 + s_{\varepsilon,i}^2)\right). \quad \blacksquare$$

C.7 Firm-Level Utilization (Proposition ??)

Firm-level utilization: $\bar{u} = \sum_i \omega_i^* u_i^{*,ss}(D_i) = \tilde{\Lambda} \sum_i \omega_i^* D_i^\psi M_i^{-1/\gamma}$.

Substituting $\omega_i^* = M_i^{1/\phi}/S$:

$$\bar{u} = \frac{\tilde{\Lambda}}{S} \sum_i D_i^\psi M_i^{1/\phi - 1/\gamma}.$$

Using $1/\phi - 1/\gamma = (1 - \psi)/\phi$ (since $\psi/\phi = 1/\gamma$):

$$\bar{u} = \frac{\tilde{\Lambda}}{S} \sum_i D_i^\psi M_i^{(1-\psi)/\phi}.$$

Taking expectations:

$$\mathbb{E}[\bar{u}] = \frac{\tilde{\Lambda}}{S} \sum_i M_i^{(1-\psi)/\phi} \mathbb{E}[D_i^\psi].$$

Firm-plus-idiosyncratic case. Using $s_{D,i}^2 = s_f^2 + s_{\varepsilon,i}^2$:

$$M_i^{(1-\psi)/\phi} \mathbb{E}[D_i^\psi] = \exp\left[\mu_i + \frac{1}{2}(\phi - (\gamma - 1)\psi^2)(s_f^2 + s_{\varepsilon,i}^2)\right]$$

where the σ^2 coefficient simplifies using $\phi = \gamma\psi$:

$$\phi(1 - \psi) + \psi^2 = \gamma\psi(1 - \psi) + \psi^2 = \phi - (\gamma - 1)\psi^2.$$

Denominator.

$$S = \sum_j \exp\left(\mu_j + \frac{1}{2}\phi(s_f^2 + s_{\varepsilon,j}^2)\right).$$

Separating the firm-shock factor. Factor $\exp(\frac{1}{2}(\phi - (\gamma - 1)\psi^2)s_f^2)$ out of the numerator and $\exp(\frac{1}{2}\phi s_f^2)$ out of the denominator:

$$\frac{\text{Num}}{\text{Den}} = \frac{\exp[\frac{1}{2}(\phi - (\gamma - 1)\psi^2)s_f^2]}{\exp[\frac{1}{2}\phi s_f^2]} \cdot \frac{\sum_i \exp[\mu_i + \frac{1}{2}(\phi - (\gamma - 1)\psi^2)s_{\varepsilon,i}^2]}{\sum_j \exp[\mu_j + \frac{1}{2}\phi s_{\varepsilon,j}^2]}.$$

The ratio of s_f^2 terms simplifies to $\exp(-\frac{1}{2}(\gamma - 1)\psi^2 s_f^2)$. Combining:

$$\mathbb{E}[\bar{u}] = \tilde{\Lambda} \cdot \exp\left(-\frac{1}{2}(\gamma - 1)\psi^2 s_f^2\right) \cdot \frac{\sum_i \exp\left[\mu_i + \frac{1}{2}(\phi - (\gamma - 1)\psi^2)s_{\varepsilon,i}^2\right]}{\sum_j \exp\left[\mu_j + \frac{1}{2}\phi s_{\varepsilon,j}^2\right]}. \quad \blacksquare$$